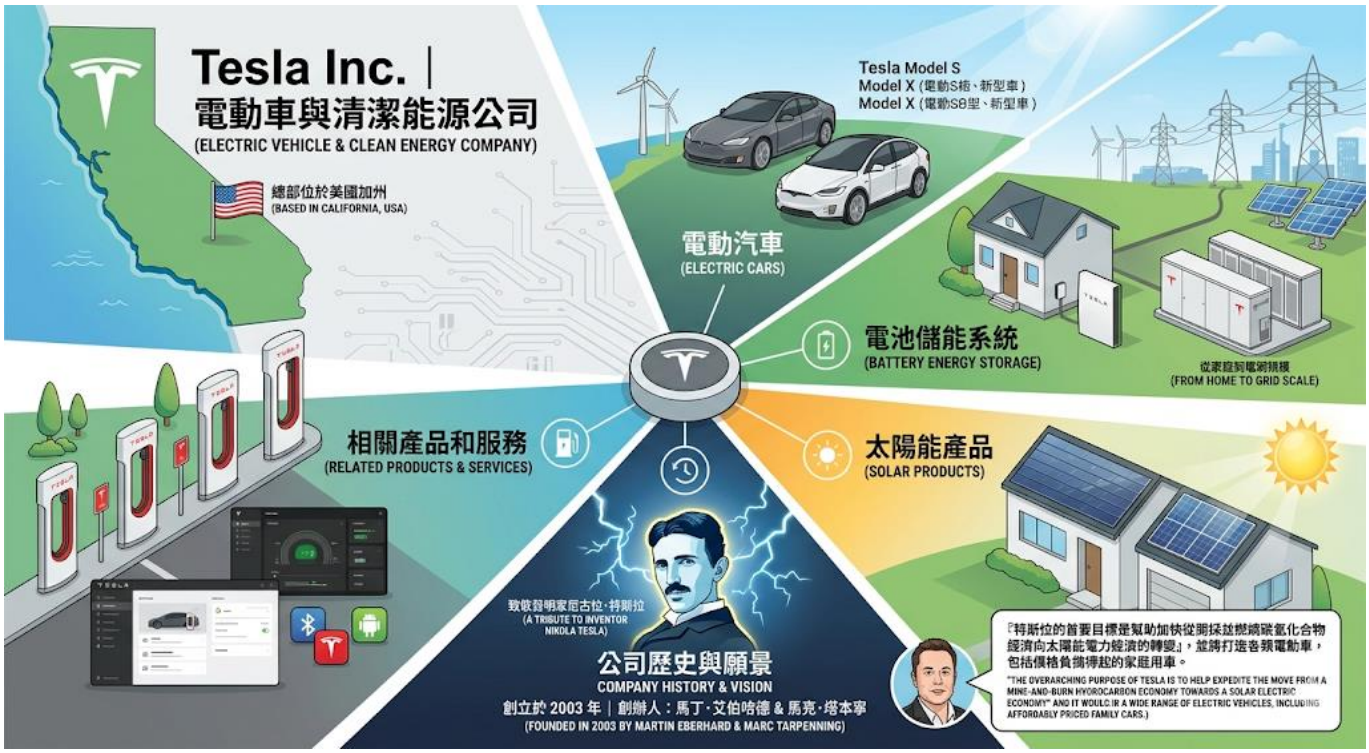
Ch2 Tesla

Tesla Inc. is an **electric vehicle** and **clean energy** company based in California, USA. Tesla's products include electric cars, **battery** energy storage from home to **grid scale**, **solar** products and related products and services. Founded in 2003 by Martin Eberhard and Marc Tarpenning, the company's name is a **tribute** to **inventor** and **electrical** engineer Nikola Tesla. Elon Musk, who formerly served as chairman and is the current **CEO**, said that "the **overarching** purpose of Tesla is to help **expedite** the move from a mine-and-burn **hydrocarbon** economy towards a **solar** electric economy" and it would build a wide range of electric vehicles, including **affordably** priced family cars.

特斯拉公司（Tesla Inc.）是一家總部位於美國加州的**電動車**與**清潔能源**公司。特斯拉的產品包括電動汽車、從家庭到**電網規模**的**電池**儲能系統、**太陽能**（**與太陽有關的**）產品以及相關產品和服務。該公司於 2003 年由馬丁·艾伯哈德（Martin Eberhard）和馬克·塔本寧（Marc Tarpenning）創立，公司名稱是為了**致敬**（**表達敬意的頌詞或禮物**）**發明家**與**電氣**工程師尼古拉·特斯拉（Nikola Tesla）。曾任董事長且現任**執行長**（**CEO**）的伊隆·馬斯克（Elon Musk）表示，「特斯拉的**首要**（**支配一切的**）目標是幫助**加快**（**促進**）從開採並燃燒**碳氫化合物**經濟向**太陽能**電力經濟的轉變」，並將打造各類電動車，包括價格**負擔得起的**家庭用車。

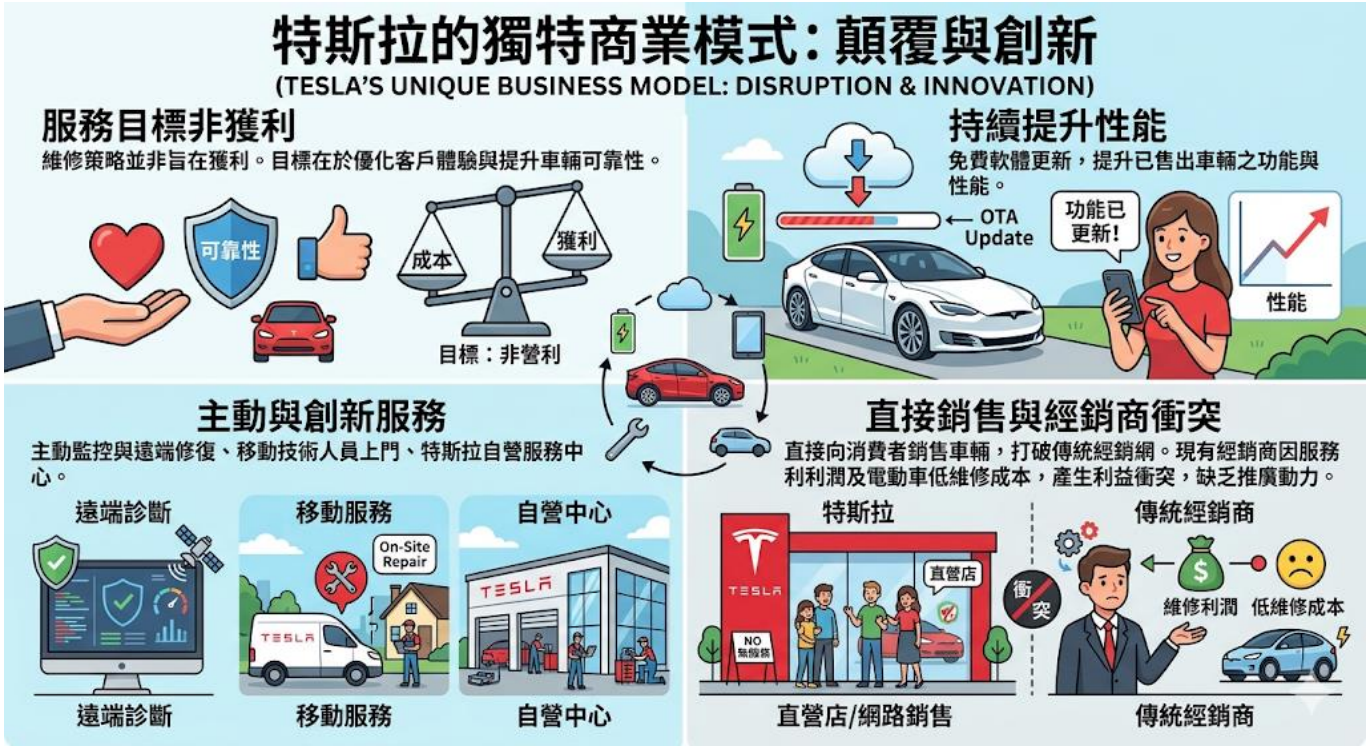


With the Model S, Tesla started to build an integrated computer hardware and software **architecture** at the center of its vehicles. Musk **indicated** that Tesla is a software company as much as it is a hardware company. Tesla provides online **software update** to its cars.

隨著 Model S 的推出，特斯拉開始在車輛中心建立整合的電腦硬體與軟體**架構**。馬斯克**表明**（**顯示**）特斯拉既是一家硬體公司，也是一家軟體公司。特斯拉為其車輛提供在線**軟體更新**。

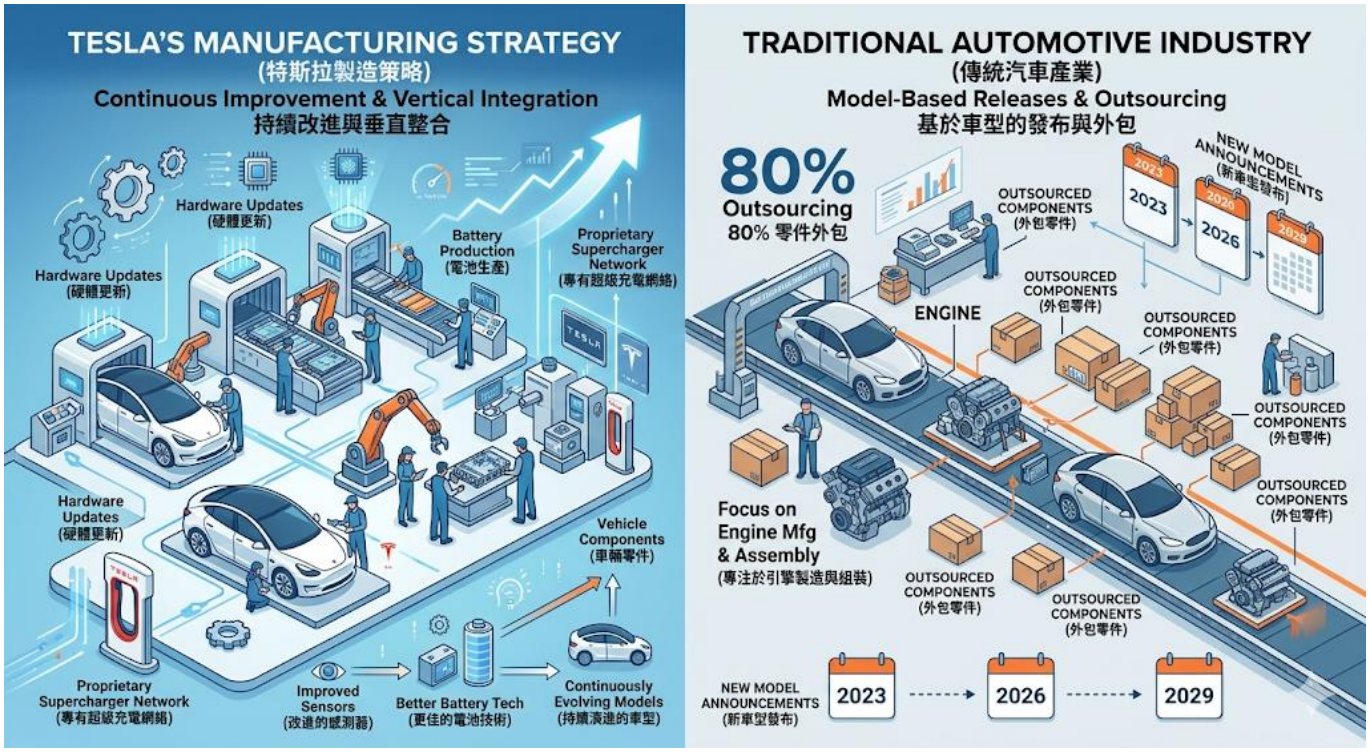
This allows improvement of the **functionality** and **performance** of the cars already-sold for free. Tesla is the first **automaker** that sells cars directly to **consumers**. All others use independently owned **dealerships**. Musk believes existing **dealerships** have a **conflict of interest** and will not **promote** electric cars from Tesla or any manufacturer, because they make more money servicing than selling cars, and electric cars have lower servicing costs. For this reason, Tesla's sell its vehicles online rather than through a conventional dealer network. The service strategy of Tesla is to service its vehicles through **proactive** monitoring, **remote diagnosis** and repair, mobile technicians, and Tesla-owned service centers. Tesla's goal is not to make a profit on service.

這讓已售出的汽車能免費提升其**功能**與**性能**（**表現**）。特斯拉是第一家直接向**消費者**銷售汽車的**汽車製造商**。其他所有廠商都使用獨立擁有的**經銷公司**。馬斯克認為現有的**經銷商**存在**利益衝突**，不會**推廣**（**促銷**）特斯拉或任何製造商的電動車，因為他們從維修服務中賺取的利潤高於賣車，而電動車的維修成本較低。因此，特斯拉通過網絡銷售車輛，而非傳統的經銷網絡。特斯拉的服務策略是通過**主動的**監控、**遠端診斷**與修理、移動技術人員以及特斯拉自營的服務中心來提供售後服務。特斯拉的目標並非從維修服務中獲利。



Tesla’s manufacturing strategy is to continuously improve the hardware of its cars, rather than waiting for announcement of a new model. Tesla’s production strategy includes a high degree of **vertical integration** which includes producing vehicle **components** as well as building **proprietary** stations where customers can **charge** Tesla vehicles. **Vertical integration** is rare in the **automotive** industry, where companies typically **outsource** 80% of **components** to suppliers, and focus on engine manufacturing and final **assembly**.

特斯拉的製造策略是持續改進車輛硬體，而非等待新車型的發布。特斯拉的生產策略包含高度的**垂直整合**，這包括生產車輛**零件**（**組成部分**），以及建立讓客戶可以為特斯拉車輛**充電**的**所有**（人）**的**（**專有的**）充電站。**垂直整合**在**自動化**（**汽車**）行業中非常罕見，該行業的公司通常將 80% 的**零件**（**成分**）**外包**（**委外**）給供應商，並專注於引擎製造和最終**組裝**（**組合體**）。

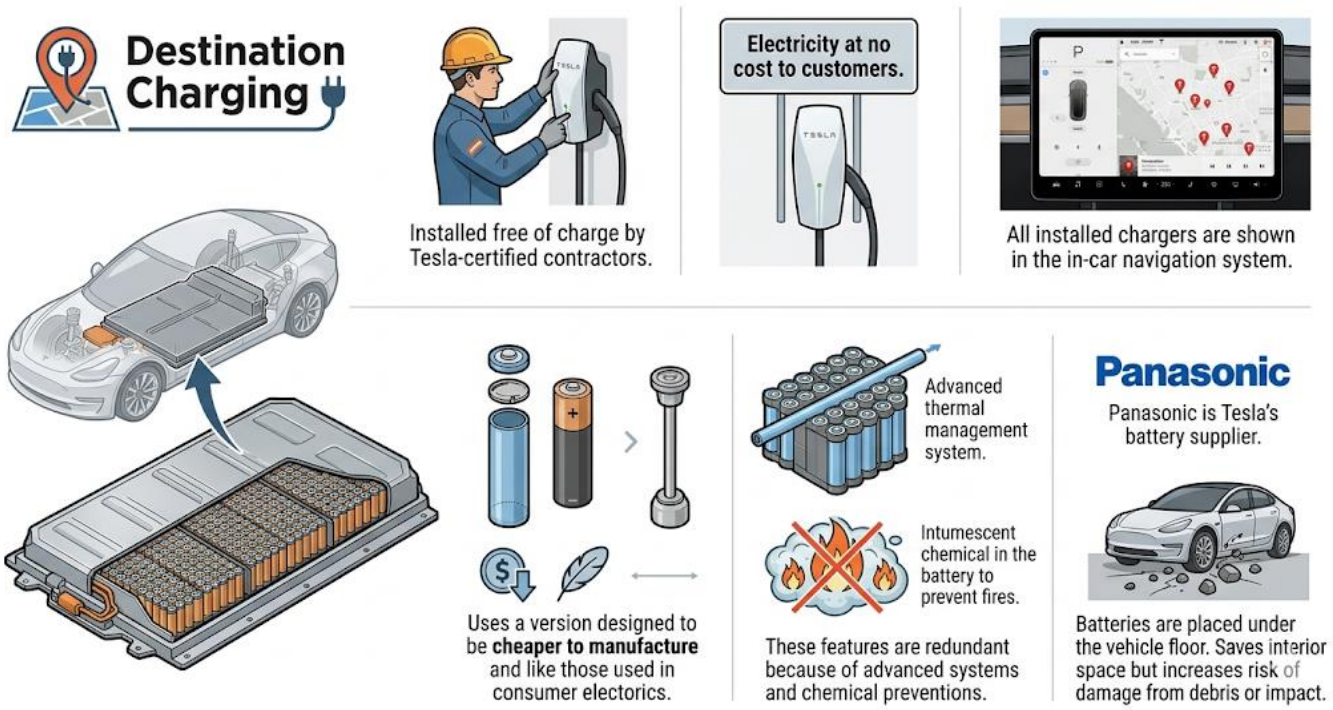


In 2014, Tesla **discreetly launched** its charging network by providing **chargers** to hotels, restaurants, shopping centers and other full service stations to provide **on-site** vehicle charging at twice the power of a typical home charging station.

2014 年，特斯拉**審慎地**（**謹慎地**）**啟動**（**推出**）了其充電網絡，向酒店、餐廳、購物中心和其他全服務站點提供**充電器**，以提供比一般家用充電站功率高出兩倍的**現場**車輛充電。

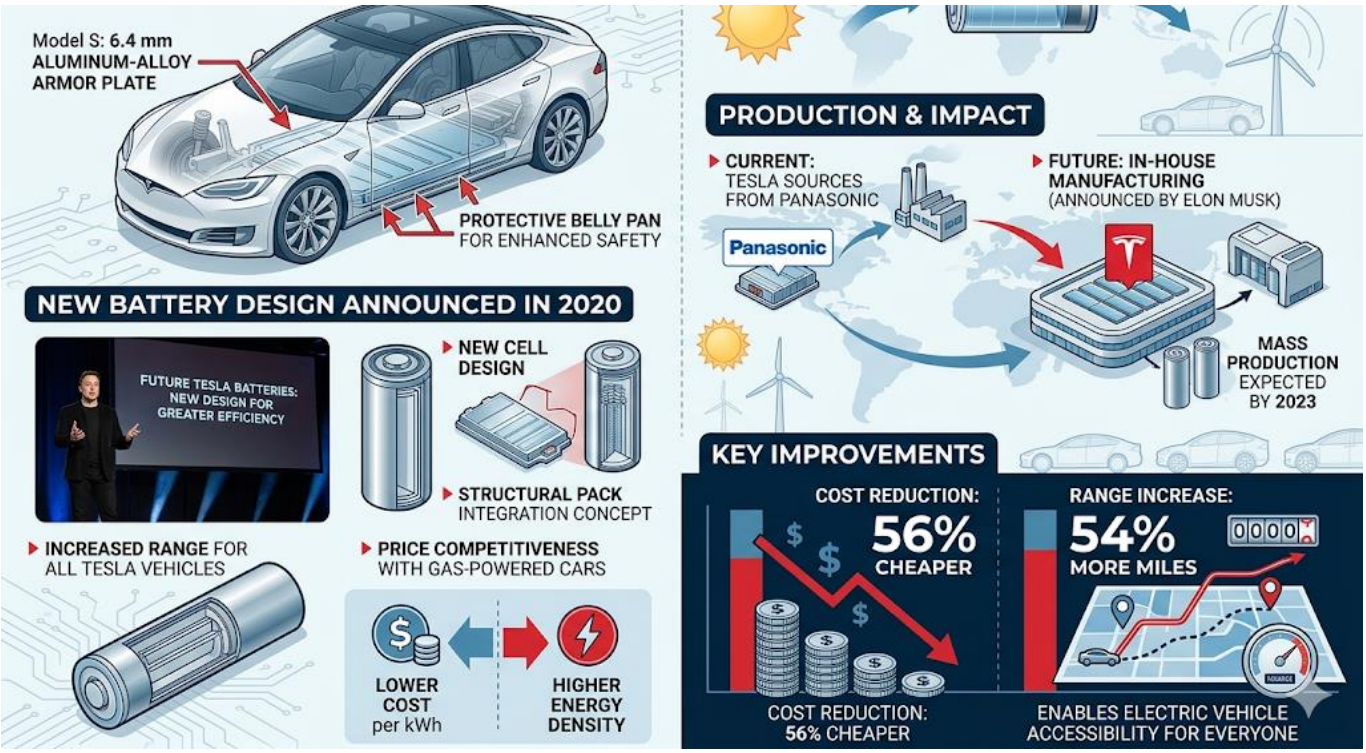
Destination **chargers** are installed **free of charge** by Tesla-**certified** contractors. The locations must provide the electricity at no cost to their customers. All installed **chargers** are shown in the in-car **navigation** system. Unlike other automakers, Tesla does not use individual large battery **cells**, but thousands of small, **cylindrical**, **lithium-ion commodity cells** like those used in consumer electronics. Tesla uses a version of these **cells** that is designed to be cheaper to manufacture and lighter than standard **cells** by removing some safety **features**. According to Tesla, these **features** are **redundant** because of the advanced thermal management system and an **intumescent** chemical in the battery to prevent fires. Panasonic is Tesla's battery supplier. The batteries of Tesla cars are placed under the vehicle **floor**. This saves **interior** and **trunk** space but increases the risk of battery damage by **debris** or impact.

目的地**充電器**由特斯拉**認證的**承包商**免費**安裝。這些場所必須免費為客戶提供電力。所有已安裝的**充電器**都會顯示在車載**導航**系統中。與其他汽車製造商不同，特斯拉不使用單個大電池，而是使用數千個小型的**柱形的**、**鋰離子商品**（**貨物**）**電池**，類似於消費電子產品中所使用的。特斯拉使用一種經過設計的**電池**版本，通過移除某些安全**特性**（**特點**），使其製造成本更低且重量比標準**電池**更輕。根據特斯拉的說法，由於先進的熱管理系統和電池中用於防火的**膨脹的**（**鼓起的**）化學物質，這些**特點**是**多餘的**（**不需要的**）。松下（Panasonic）是特斯拉的電池供應商。特斯拉汽車的電池放置在車輛**地板**下方。這節省了**內部**和**後車箱**空間，但也增加了電池被**殘骸**（**碎片**）或撞擊損壞的風險。



The Model S has 6.4 mm aluminum-alloy **armor plate**. In 2020, Tesla announced a new design for their future batteries. This design will increase the range of Tesla Vehicles. It will also allow future electric vehicles prices to compete with their gas-powered **counterparts**. Currently, Tesla **sources** their batteries from Panasonic, however with the new design of the batteries Elon Musk has announced plans to manufacture the batteries **in-house**. The new batteries are expected to be **mass-produced** by 2023. They will be 56% cheaper and allow the cars to travel 54% more miles.

Model S 配備了 6.4 毫米厚的鋁合金**裝甲板**。2020 年，特斯拉宣布了未來電池的新設計。這項設計將增加特斯拉車輛的續航里程。它還將使未來的電動車價格能與燃油車**對應產品**（**作用相同者**）競爭。目前，特斯拉從松下**獲取**（**來源**）電池，然而隨著電池新設計的出現，馬斯克已宣布計劃在公司**內部**（**自行生產**）製造電池。新電池預計將於 2023 年開始**大量生產**。它們的成本將降低 56%，並使汽車的行駛里程增加 54%。



Autopilot is an important **feature** of Tesla cars in addition to **clean energy**. **Autopilot** is designed to assist drivers with the most **burdensome** parts of driving in order to make Tesla cars safer and more capable over time. Tesla states that current **autopilot features** require active driver **supervision** and do not make the vehicle **autonomous**. Starting 2014, all Tesla cars are **shipped** with **sensors** and software to support **autopilot**. Tesla upgraded its **sensors** and software in 2016 to support full **self-driving**.

除了**清潔能源**外，**自動駕駛輔助系統**（**Autopilot**）是特斯拉汽車的一項重要**特色**。**自動駕駛系統**旨在協助駕駛應對駕駛中最**繁重的**（**麻煩的**）部分，以便讓特斯拉汽車隨著時間推移變得更安全且功能更強。特斯拉聲明，目前的**自動駕駛****功能**仍需要駕駛的主動**監督**（**管理**），並未使車輛達到完全的**自主**（**自治的**）。從 2014 年開始，所有特斯拉汽車在**運送**（**運輸**）時皆配備了**感測器**與軟體以支援**自動駕駛**。特斯拉在 2016 年升級了其**感測器**與軟體，以支持全**自動駕駛**。

The system includes eight cameras, twelve **ultrasonic sensors**, and forward-facing radar. Musk predicted that drivers would be able to sleep in their vehicle while it drives itself. In April 2019, Tesla announced that all of its cars will include **Autopilot** software as a standard **feature** moving forward. Full **self-driving** software is an extra cost option. On April 24, 2020, Tesla released a software update to its full **self-driving** capability. With this update, cars **recognize** and automatically stop at stop signs. The cars also automatically slow down and eventually stop at traffic lights until the driver has **indicated** that it is safe to proceed. Tesla acknowledges that the software is still in a **beta phase** and far from being finished.
該系統包括八個攝像頭、十二個**超聲的****感測器**以及前向雷達。馬斯克預測，駕駛將能夠在車輛自動行駛時在車內睡覺。2019 年 4 月，特斯拉宣布今後所有車輛都將把**自動駕駛**軟體列為標準**功能**（**特色**）。全**自動駕駛**軟體則是一項額外付費選項。2020 年 4 月 24 日，特斯拉發布了針對全**自動駕駛**能力的軟體更新。透過這次更新，車輛能**識別**（**認出**）並在停車標誌前自動停車。車輛還會自動減速並最終在紅綠燈前停車，直到駕駛**表明**（**指示**）可以安全行進。特斯拉承認該軟體仍處於 **Beta 階段**（**測試階段**），且遠未完成。

affordable (adj): 負擔得起的
architecture (n): 架構
assembly (n): 組合體; 組成
auto (n): 汽車
auto- (prefix): 自己的; 本身的; 自動的
automaker (n): 汽車製造商
automative (adj): 自動化的
autonomous (adj): 自治的, 有自治權的
autopilot (n): 自動駕駛輔助系統
battery (n): 電池
beta phase (n): 測試階段; Beta 階段
burdensome (adj): 麻煩的; 沉重的, 繁重的
cell (n): 電池/細胞
CEO (n): 執行長
certified (adj): 有合格證書的; 獲得資格的
charge (v): 充電
charger (n): 充電器, 充電機
clean energy (n): 清潔能源
commodity (n): 商品, 貨物
component (n): 組成部分; 成分; 零件
conflict of interest (n): 利益衝突
consumer (n): 消費者
counterpart (n): 作用相同者, 相對應者
cylindrical (adj): 柱形的
debris (n): 碎片, 殘骸
dealership (n): (獲准出售特定產品的) 經銷公司
diagnosis (n): 診斷
discreetly (adv): 審慎地, 謹慎地, 小心地
electric vehicle (n): 電動車
electrical (adj): 電的; 與電有關的
expedite (v): 迅速完成; 促進; 加快
feature (n): 特色, 特徵, 特點
floor (n): (房間的) 地板, 地面
free of charge (phr): 免費
functionality (n): 功能
grid (n): 網格, 方格; 格柵, 格子
grid scale (n): 電網規模
hydrocarbon (n): 碳氫化合物
in-house (adv): 自行 (設計、生產)
indicate (v): 標示; 表明; 顯示; 暗示
integrate (v): (使) 融入; (使) 成為一體
interior (n): 內部
intumescent (adj): 膨脹的, 鼓起的
ion (n): 離子
inventor (n): 發明家
launch (v): 啟動, 推出, 發起
lithium (n): 鋰
mass-produce (v): 大量生產
navigation (n): 領航; 導航
on-site (adv): 在工地的; 在現場的
outsource (v): 外包, 委外
overarching (adj): 首要的; 包羅萬象的, 支配一切的
performance (n): 性能; 表現
proactive (adj): 主動的, 積極的
promote (v): 促進; 促銷, 推銷; 推廣
proprietary (adj): 所有 (人) 的, 業主的
recognize (v): 認出, 認識; 識別
redundant (adj): 多餘的, 不需要的; 累贅的, 囉唆的
remote diagnosis (n): 遠端診斷
self-driving (adj): 自動駕駛的
sensor (n): 感測器
ship (v): 運輸, 運送
software update (n): 軟體更新
solar (adj): 太陽能的; 與太陽有關的
source (n/v): 來源/從 (某地) 獲取
supervision (n): 監督, 管理, 指導
tribute (n): (尤指在正式場合表達敬意的) 頌詞, 禮物
truck (n): 卡車/後車箱
ultrasonic (adj): (聲音) 超聲的
vertical integration (n): 垂直整合